

GPS Vehicle Status Logic Explained

This document explains the recommended two-step vehicle status engine used in GPS tracking systems.

Step 1: Check GPS Freshness

Before determining whether a vehicle is Running, Idle, Moving, or Parked, first determine how old the latest GPS update is. If GPS data is not recent, motion information cannot be trusted.

Time Since Last GPS	Displayed Status	Meaning
Under 2 minutes	Live	GPS data is current; calculate motion status.
2–10 minutes	Delayed	GPS updates are late; keep previous motion in background.
10–30 minutes	Weak Signal / Stale	Tracker has not reported recently.
Over 30 minutes	Offline	No GPS report for over 30 minutes. This does NOT mean the

Important Note

Offline means no GPS communication has been received for more than 30 minutes. It does not mean the engine is off. A vehicle can still be running but appear Offline if the tracker loses network connectivity or the SIM card is removed.

Step 2: If GPS is Live (<2 minutes), determine status using Ignition and Speed

Ignition	Speed	Status	Explanation
ON	>1 km/h	Running	Engine on and vehicle moving.
ON	≤1 km/h	Idle	Engine on but vehicle stationary (traffic, waiting, loading).
OFF	>1 km/h	Moving	Vehicle moving with ignition off (towed, rolling, wiring issue).
OFF	≤1 km/h	Parked	Engine off and vehicle stationary.

Decision Flow

1. Receive a GPS update.
2. Check how old the latest GPS report is.
3. If older than 30 minutes → Offline.
4. If 10–30 minutes → Weak Signal / Stale.
5. If 2–10 minutes → Delayed.
6. If under 2 minutes → Use Ignition and Speed:
 - ON + Moving → Running
 - ON + Stopped → Idle
 - OFF + Moving → Moving
 - OFF + Stopped → Parked

Example Scenarios

GPS Age	Ignition	Speed	Result
30 sec	ON	60 km/h	Running
45 sec	ON	0 km/h	Idle
20 sec	OFF	25 km/h	Moving
40 sec	OFF	0 km/h	Parked
5 min	ON	70 km/h	Delayed
20 min	OFF	0 km/h	Weak Signal / Stale
45 min	ON	80 km/h	Offline (data too old)